

DIGITAL FORENSICS INVESTIGATION REPORT

1. Case Overview

This report summarizes findings from a forensic analysis of a disk image related to suspected data exfiltration and ransomware infection. The investigation focuses on identifying stolen information, involved individuals, malware presence, and potential recovery options for affected data.

Before conducting any work the image received was hashed using sha1sum:

```
b4c3ae80f840bb612f982ba5081872b8a6a19e83  forensicImage.7z
```

```
ec62a619f0252ed89acda43616f3a267bfa3f650  DigitalForensic_AssignmentImage Clone.qcow
```

Then I converted the image to raw data using qemu-img so autopsy could parse it better:

```
b4c3ae80f840bb612f982ba5081872b8a6a19e83  forensicImage.7z
```

```
ec62a619f0252ed89acda43616f3a267bfa3f650  DigitalForensic_AssignmentImage Clone.qcow
```

Then I was ready to jump into analysis.

2. Analysis

2.1 What information was stolen?

Evidence indicates that **design concepts related to a new game developed by Sabrina** were unlawfully accessed and shared. The investigation found email communications suggesting that Peter forwarded confidential design material belonging to Sabrina to another individual.

2.2 Which persons were involved in the case?

The following individuals were identified:

- **Peter** – internal employee who accessed and leaked confidential information
- **Iris** – internal employee who manipulated Peter using emotional techniques to make him leak designs from Sabrina

Analyzing email artifacts found in Users\Peter\AppData\Roaming\Peter1983\sent is clear that Peter leaked the information to Iris. Emails found that support this statement:

Thread 1

Sure, let's discuss the details at work. :)

> Yeah no problem at all.

> Say, do you want to go for a drink someday? ;)

> On 2016-08-24 01:28, Peter wrote:

> > Hey,

> > nice! Puh the traffic today was terrible...

> > Hope you had a nice ride. :)

> > Am Wed, 24 Aug 2016 00:48:30 +0200

> > schrieb briennefan@openmailbox.org:

> >> Hihi

> >> Hi Peter, now we can chat. ;)

Thread 2

Sure thing :)

Am Wed, 24 Aug 2016 01:40:15 +0200

schrieb briennefan@openmailbox.org:

> I'm ok with that :)

> I also think, that it was a date :)

> But it should be a thing between us two and we should keep it as a

> secret at work. ;)

> On 2016-08-24 01:36, Peter wrote:

> > Hey,

> > it felt like one, hope do not take it wrong, that I call our

> > meeting a date.

Thread 3

Uhm ok, I'll look what I can do for you :)

Am Wed, 24 Aug 2016 01:51:15 +0200

schrieb briennefan@openmailbox.org:

> I'm very impatient. Please do it for me, maybe I will reward you with

> another date? ;)

> On 2016-08-24 01:49, Peter wrote:

> > Hehe why don't you just wait until she presents the first 3d model?

> > Sometimes she let her drawings unlocked on her table, but I don't

> > think, that I should copy them :/
> > Am Wed, 24 Aug 2016 01:46:58 +0200
> > schrieb briennefan@openmailbox.org:
> >> Can you get a copy for me? I'm very interested in it :)
> >> On 2016-08-24 01:45, Peter wrote:
> >> > Nope, not really. She only shows it to anna and john, but I know,
> >> > that she keeps it in her desk. Why?
> >> > Am Wed, 24 Aug 2016 01:43:05 +0200
> >> > schrieb briennefan@openmailbox.org:
> >> >> Hey, can you do me a favor?
> >> >> Have you seen some design concepts of sabrina?

Thread 4

Hey did I do something wrong? You've been acting strange lately... :(

Thread 5

Hey Iris!

Anna and john asked me today, if I leaked some information about our work. I denied everything, I don't want you to get into trouble. What have you done with the desired item from Sabrina. Please answer me
Iris, I don't want to discuss this at Work.

Peter

These messages reveal a personal relationship between Peter and Iris, how she requested confidential design materials, and how Peter ultimately leaked them.

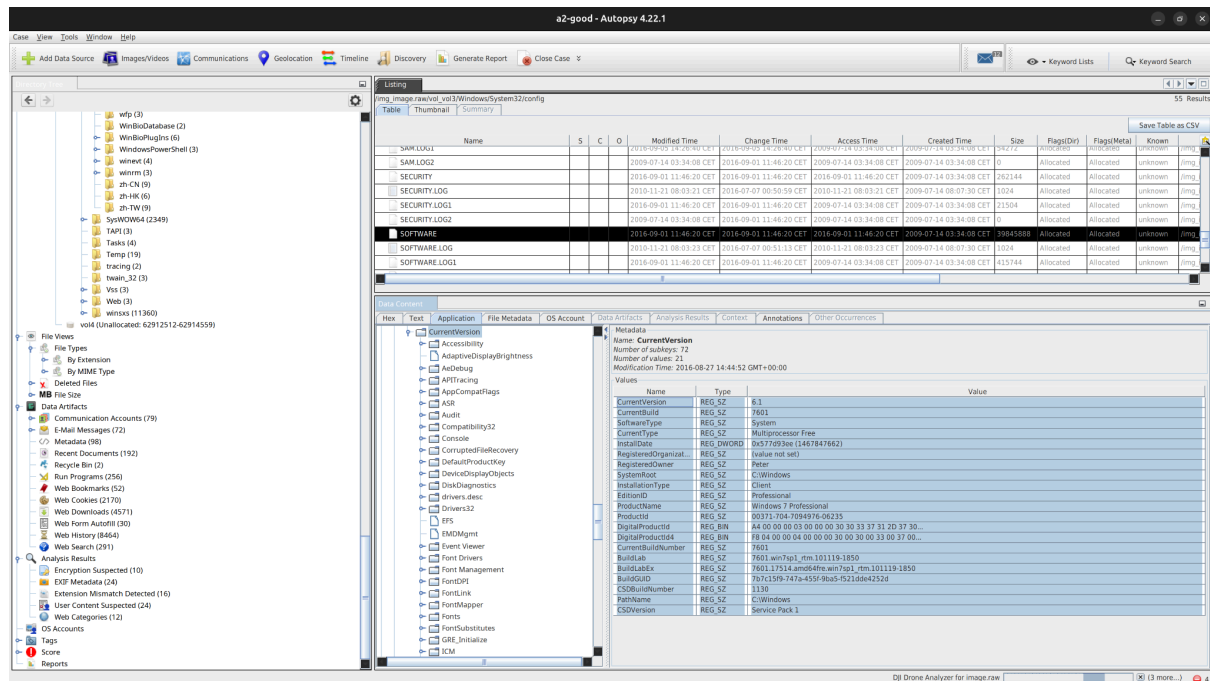
2.3 Is there any further information that may be helpful regarding the ongoing investigations?

It would be useful to analyze Iris's computer to gain further insight into how the information was leaked and to support potential legal action.

3. Technical Analysis

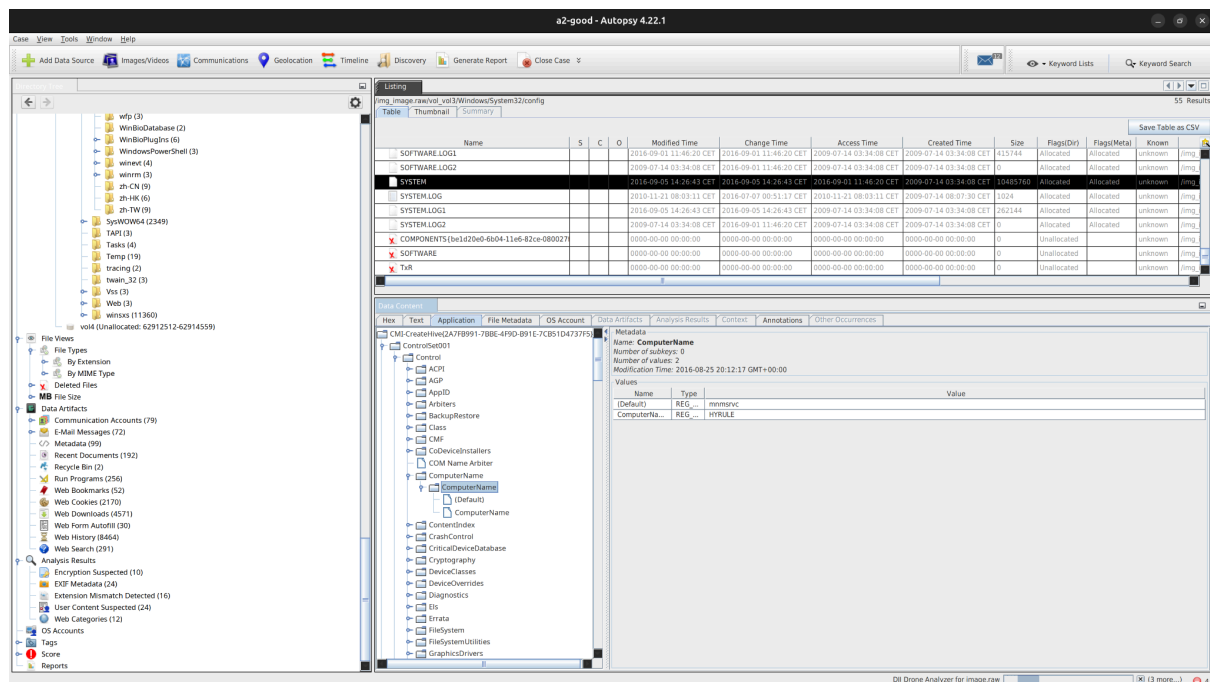
3.1 What operating system was used?

Microsoft Windows 7 Professional



3.2 What is the computer's name?

Hyrule



3.3 When was the operating system installed, when was it running the last time?

The operating system was installed on July 6, 2016.

Determining the last time the system was running is more complex. Although the system records the last shutdown time, this value does not guarantee accuracy, since a sudden loss of power or a system crash can cause the computer to shut down without properly updating the timestamp.

Therefore, we can analyze the last execution of `sppsvc.exe`, the process responsible for validating the Microsoft license at boot time. This value can help determine the last time the system was booted. However, this alone is not sufficient, as the computer could have remained powered on for an extended period after startup.

The final method used to estimate the last system activity is to examine the last executed programs. In this case, it matches the last execution of `sppsvc.exe`, indicating activity on October 5, 2016.

Case **Xim** Tools **Window** **Help**

+ Add Data Source
 I Images/Videos
 C Communications
 L Geolocation
 T Timeline
 D Discovery
 G Generate Report
 X Close Case

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F File Views

- F File Types
- E By Extension
- B By MIME Type
- X Deleted files
- MB File Size

D Data Artifacts

- C Communication Accounts (11333)
- C Contacts (1)
- E E-Mail Messages (6038)
- M Metadata (1751)
- R Recent Documents (192)
- R Recycle Bin (2)
- B Run Programs (266)
- B Web Bookmarks (52)
- C Web Cookies (2170)
- W Web Downloads (4971)
- W Web Form Autofill (30)
- W Web History (8464)
- W Web Search (291)

L Listing

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R Run Programs

T Table
 Th Thumbnail
 S Summary

Source Name	S	C	O	Program Name	Path	Date/Time	Count	Comment	Data Source
✓ WERNGR.EXE-0F2AC8BC.pf				WERNGR.EXE	WINDDOWS\SYSTEM32	2016-09-05 00:07:50 CET	15	Prefetch File	image.raw
✓ CONHOST.EXE-1F3E0E10.pf				CONHOST.EXE	WINDDOWS\SYSTEM32	2016-09-05 00:07:50 CET	31	Prefetch File	image.raw
✓ TASKHOST.EXE-723F8F31D.pf				TASKHOST.EXE	WINDDOWS\SYSTEM32	2016-09-05 00:07:50 CET	35	Prefetch File	image.raw
✓ WERNGR.EXE-0F2AC8BC.pf				WERNGR.EXE	WINDDOWS\SYSTEM32	2016-09-05 00:07:50 CET	15	Prefetch File	image.raw
✓ CONHOST.EXE-1F3E0E10.pf				CONHOST.EXE	WINDDOWS\SYSTEM32	2016-09-05 00:07:50 CET	31	Prefetch File	image.raw
✓ TASKHOST.EXE-723F8F31D.pf				TASKHOST.EXE	WINDDOWS\SYSTEM32	2016-09-05 00:07:50 CET	35	Prefetch File	image.raw
✓ WHMPVSE.EXE-1620551C.pf				WHMPVSE.EXE	WINDDOWS\SYSTEM32\WBEM	2016-09-05 14:27:27 CET	60	Prefetch File	image.raw
✓ WHMPVSE.EXE-1620551C.pf				WHMPVSE.EXE	WINDDOWS\SYSTEM32\WBEM	2016-09-05 14:27:27 CET	60	Prefetch File	image.raw
✓ DLHUST.EXE-40D0444D.pf				DLHUST.EXE	WINDDOWS\SYSTEM32	2016-09-05 14:28:09 CET	47	Prefetch File	image.raw
✓ DLHUST.EXE-40D0444D.pf				DLHUST.EXE	WINDDOWS\SYSTEM32	2016-09-05 14:28:09 CET	47	Prefetch File	image.raw
✓ SVCHOST.EXE-05F624AB.pf				SVCHOST.EXE	WINDDOWS\SYSTEM32	2016-09-05 14:28:43 CET	38	Prefetch File	image.raw
✓ SPSPVC.EXE-B0F8131B.pf				SPSPVC.EXE	WINDDOWS\SYSTEM32	2016-09-05 14:28:43 CET	37	Prefetch File	image.raw
✓ SVCHOST.EXE-05F624AB.pf				SVCHOST.EXE	WINDDOWS\SYSTEM32	2016-09-05 14:28:43 CET	38	Prefetch File	image.raw
✓ SPSPVC.EXE-B0F8131B.pf				SPSPVC.EXE	WINDDOWS\SYSTEM32	2016-09-05 14:28:43 CET	37	Prefetch File	image.raw

Hex Text Application
Source File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

Results: 2 of 2
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Type	Value	Source(s)
Program Name	SPSPVC.EXE	Windows Prefetch Analyzer
Path	WINDDOWS\SYSTEM32	Windows Prefetch Analyzer
Date/Time	2016-09-05 14:28:43 CET	Windows Prefetch Analyzer
Count	37	Windows Prefetch Analyzer
Comment	Prefetch File	Windows Prefetch Analyzer
Source File Path	jimg_image.raw\vol_003\Windows\Prefetch\SPSPVC.EXE-B0F8131B.pf	Windows Prefetch Analyzer
Artifact ID	9223372036854754919	

F File Views

- F File Types
- E By Extension
- B By MIME Type
- X Deleted files
- MB File Size

D Data Artifacts

- C Communication Accounts (11333)
- C Contacts (1)
- E E-Mail Messages (6038)
- M Metadata (1751)
- R Recent Documents (192)
- R Recycle Bin (2)
- B Run Programs (266)
- B Web Bookmarks (52)
- C Web Cookies (2170)
- W Web Downloads (4971)
- W Web Form Autofill (30)
- W Web History (8464)
- W Web Search (291)

A Analysis Results

- E Encryption Suspected (10)
- E EXIF Metadata (52)
- E Extension Mismatch Detected (19)
- I Interesting Items (1)
- U User Content Suspected (52)
- W

3.4 What is the SID (Security Identifier) for the user Peter?

S-1-5-21-3030563450-629482277-3432984521-1003

The screenshot shows the Autopsy 4.22.1 interface. On the left is a file system tree. The center pane displays a table of system logs. The right pane shows a detailed view of a specific profile.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
SECURITY.LOG				2010-11-21 08:03:21 CET	2010-07-07 00:50:59 CET	2010-11-21 08:03:21 CET	2009-07-14 08:07:30 CET	1024	Allocated	Allocated	unknown
SECURITY.LOG1				2010-09-01 11:46:20 CET	2010-09-01 11:46:20 CET	2009-07-14 03:34:08 CET	2009-07-14 03:34:08 CET	21304	Allocated	Allocated	unknown
SECURITY.LOG2				2009-07-14 03:34:08 CET	2010-09-01 11:46:20 CET	2009-07-14 03:34:08 CET	2009-07-14 03:34:08 CET	0	Allocated	Allocated	unknown
SOFTWARE				2010-09-01 11:46:20 CET	2010-09-01 11:46:20 CET	2010-09-01 11:46:20 CET	2009-07-14 03:34:08 CET	39845888	Allocated	Allocated	unknown
SOFTWARE.LOG				2010-11-21 08:03:21 CET	2010-07-07 00:51:13 CET	2010-11-21 08:03:21 CET	2009-07-14 08:07:30 CET	1024	Allocated	Allocated	unknown
SOFTWARE.LOG1				2010-09-01 11:46:20 CET	2010-09-01 11:46:20 CET	2009-07-14 03:34:08 CET	2009-07-14 03:34:08 CET	415744	Allocated	Allocated	unknown
SOFTWARE.LOG2				2009-07-14 03:34:08 CET	2010-09-01 11:46:20 CET	2009-07-14 03:34:08 CET	2009-07-14 03:34:08 CET	0	Allocated	Allocated	unknown
SYSTEM				2010-09-05 14:26:43 CET	2010-09-05 14:26:43 CET	2010-09-01 11:46:20 CET	2009-07-14 03:34:08 CET	10485760	Allocated	Allocated	unknown
SYSTEM.LOG				2010-11-21 08:03:11 CET	2010-07-07 00:51:17 CET	2010-11-21 08:03:11 CET	2009-07-14 08:07:30 CET	1024	Allocated	Allocated	unknown

The right pane shows the 'S-1-5-21-3032217210-430098460-752710606-1001' profile. The 'Values' section shows the following data:

Name	Type	Value
ProfileImagePath	REG_EXPAND_SZ	C:\Users\Peter
Flags	REG_DWORD	0x00000000 (0)
State	REG_DWORD	0x00000000 (0)
Sid	REG_BINARY	B3 05 00 00 00 00 05 15 00 00 00 7A F6 BB B4...
ProfileLoadTimeLow	REG_DWORD	0x00000000 (0)
ProfileLoadTimeHigh	REG_DWORD	0x00000000 (0)
RelCount	REG_DWORD	0x00000000 (0)
RunLogonScripts...	REG_DWORD	0x00000000 (0)

3.5 Can you find traces of malware on the system? If yes...

- What kind of malware is it and how did you find it?

I found traces of ransomware malware, the first sign was this email by Peter to tech support.

Hello Dude!

Can you help me? Suddenly my files changed to mp3's and I got some

strange messages with "IMPORTANT INFORMATION" and I cannot open those

files... I did nothing great, I browsed the web for stuff and then I

got this message...

Maybe you can help me? I got some of the data back, but I saved the mp3

to be safe.

Thank you

Peter

This led me to search for the keyword "IMPORTANT INFORMATION". That led me to find the following message:

recovery+wdbic.txt NOT YOUR LANGUAGE? USE <https://translate.google.com>

What happened to your files ?

All of your files were protected by a strong encryption with RSA4096

More information about the encryption keys using RSA4096 can be found here:
[http://en.wikipedia.org/wiki/RSA_\(cryptosystem\)](http://en.wikipedia.org/wiki/RSA_(cryptosystem))

How did this happen ?

!!! Specially for your PC was generated personal RSA4096 Key , both public and private.

!!! ALL YOUR FILES were encrypted with the public key, which has been transferred to your computer via the Internet.

!!! Decrypting of your files is only possible with the help of the private key and decrypt program , which is on our Secret Server

What do I do ?

So, there are two ways you can choose: wait for a miracle and get your price doubled, or start obtaining BITCOIN NOW! , and restore your data easy way.

If You have really valuable data, you better not waste your time, because there is no other way to get your files, except make a payment.

For more specific instructions, please visit your personal home page, there are a few different addresses pointing to your page below:

1 - <http://sondr5344ygfweyjbfbkw4fhsefv.heliofetch.at/E19054AC2D59D238>

2 - <http://pts764gt354fder34fsqw45gdfsavadsfgsfg.kraskula.com/E19054AC2D59D238>

3 - <http://yyre45dbvn2nhbefbmh.begumvelic.at/E19054AC2D59D238>

If for some reasons the addresses are not available, follow these steps:

1 - Download and install tor-browser: <http://www.torproject.org/projects/torbrowser.html.en>

2 - After a successful installation, run the browser

3 - Type in the address bar: xlowfznrg4wf7dli.onion/E19054AC2D59D238

4 - Follow the instructions on the site

IMPORTANT INFORMATION

Your personal pages

http://sondr5344ygfweyjbfbkw4fhsefv.heliofetch.at/E19054AC2D59D238

http://pts764gt354fder34fsqw45gdfsavadfgsfg.kraskula.com/E19054AC2D59D238

http://yyre45dbvn2nhbefbmh.begumvelic.at/E19054AC2D59D238

Your personal page Tor-Browser xlowfznrg4wf7dli.ONION/E19054AC2D59D23

After some investigation, I found some internet searches with the same timestamp regarding a cheat for an online video game called Clash of Clans, which is likely the source of the infection.

- Which data is affected? Is it possible to restore the affected data?

25085-contactdata.csv.mp3

81058-monster_concept_art_vii_by_d_faultx.jpg.mp3

81061-myrating.csv.mp3

81064-passwords.docx.mp3

81067-unityassetstoreguide.pdf.mp3

81103-IMG_20160823_130922.jpg.mp3

81164-Fallen_Champions_concept_art_3.jpg.mp3

81167-Fungus-Documentation.pdf.mp3

81937-companydata.csv.mp3

81940-Computing_short19.pdf.mp3

All these files with type mismatches and timestamps after the infection are the affected files, at least the ones that have not been deleted and that have been found by autopsy. A quick search in google for ransomware malware that encrypts to mp3 reveals that this virus is likely teslaCrypt. An issue from 2016 on a Microsoft forum reveals an user with the same exact message on his computer.

[https://learn.microsoft.com/en-us/answers/questions/2715968/files-encrypted-by-teslacrypt-3-0-\(-mp3-extension\)](https://learn.microsoft.com/en-us/answers/questions/2715968/files-encrypted-by-teslacrypt-3-0-(-mp3-extension))

Searching for TeslaCrypt decrypt files lands me on a webpage of Cisco Talos and a tool to decrypt the encrypted data thanks to the arrest of the original actor.

4. Conclusion

The investigation confirms:

- Unauthorized disclosure of confidential design data
- Involvement of Peter and Iris in data transfer activity
- Presence of ransomware infection consistent with TeslaCrypt-like behavior
- Significant file encryption or renaming affecting user data